

Kure (naval arsenal & BCOF headquarters)

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



呉（呉市中央）

Study area	2.4 km × 1.5 km around 34.2440° N, 132.5580° E
Oldest aerial photograph	1961–1969
Aerial series in this brief	1961–1969, 1974–1978, 1979–1983, 1987–1990, 2023
Main natural landform	Floodplain and coastal plain (34%)
Former river channels	none mapped
Filled, reclaimed or drained land	20%
Former sea, river or pond	20%
Meiji-era paddies, marsh and flats	rice paddy 12%, tidal flat or beach 6.9%, uncultivated land 3.4%
Disaster memorials within 3 km	3 (landslide or debris flow 2, earthquake 1, flood 1)

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, 1961–1969 and 2023



1961–1969 aerial photograph — GSI Tiles *ort_old10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2023 annual orthophoto — GSI Tiles *nendophoto2023*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

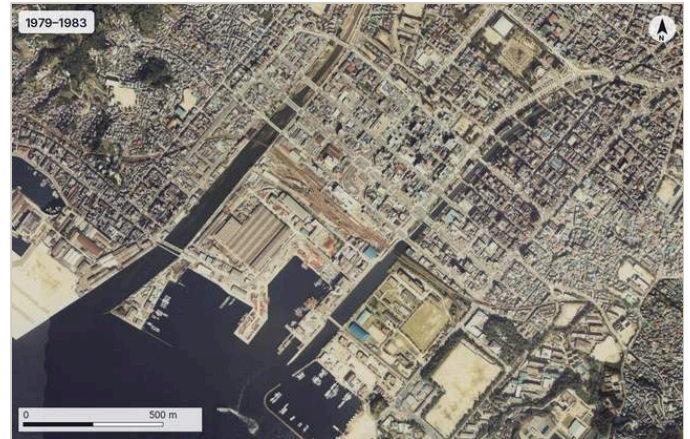
How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

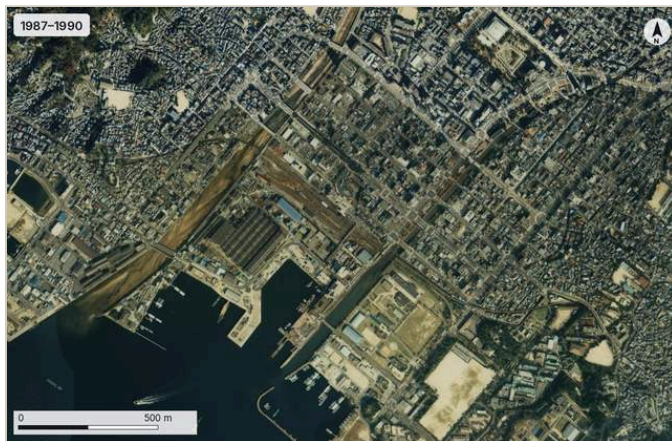
Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



1974–1978 · GSI Tiles *gazo1*, processed



1979–1983 · GSI Tiles *gazo2*, processed



1987–1990 · GSI Tiles *gazo4*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

■ Floodplain and coastal plain 34%	■ Former water area 20%
■ Mountain and hill slopes 12%	■ Alluvial fan 7.4%
■ Plateaus and terraces 5.0%	■ Water 3.6%
■ Riverbed and beach 0.8%	

What the main landforms mean

Floodplain and coastal plain (34%). Low, flat ground of little relief, built from sand and mud left by floods, or former seabed that emerged. River flooding. Ground is softer nearer the coast and shakes somewhat more; liquefaction tendency is strong on coastal plains and somewhat strong on floodplains. Storm surge near the coast.

Former water area (20%). Land shown by old maps and aerial photographs to have been sea, lake, pond, reservoir or river, later turned into land. Soft ground: very strong liquefaction tendency where filled, strong where drained. Storm surge near the coast.

Mountain and hill slopes (12%). Ridges, valleys and fairly steep ground, including terrace-scarp slopes and volcanic ground. Heavy rain or earthquakes can trigger slope failures, debris flows and landslides.

Alluvial fan (7.4%). Gently sloping ground spreading fan-like from a valley mouth, built from sediment dropped by floods. Floodwater from the mountains, and debris flows near the valley mouth. Relatively firm ground; a slight liquefaction tendency on the lower fan.

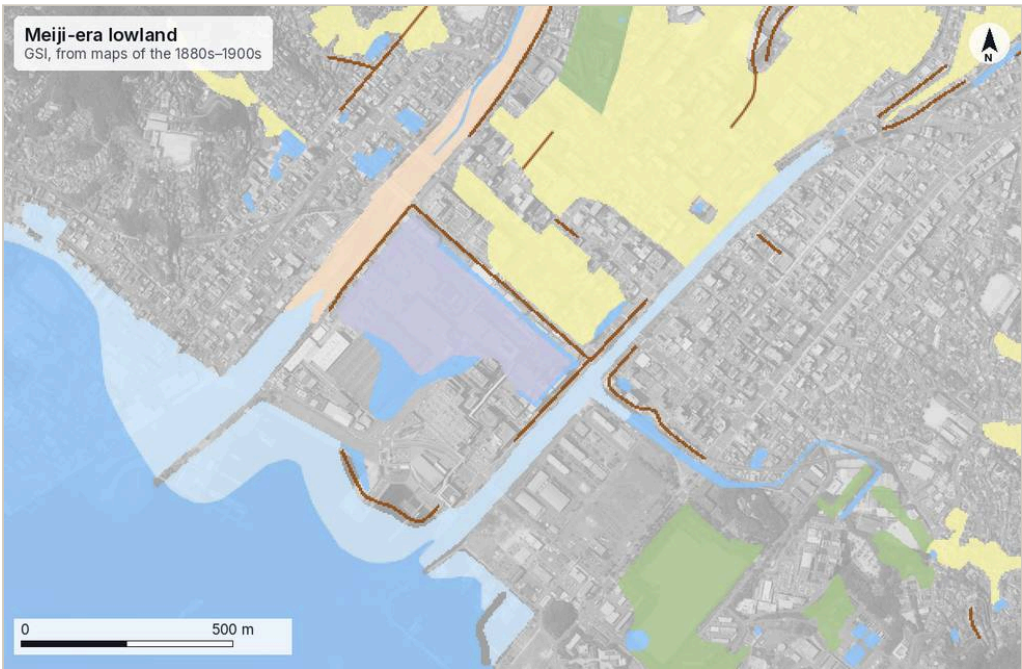
Plateaus and terraces (5.0%). Flat ground standing in steps above its surroundings, left behind as the land around it was eroded. River flooding is rarely a threat unless the terrace is only slightly above the river; watch the slopes at its edges. Firm ground: shaking is relatively weak and liquefaction is unlikely.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

Man-made ground

- Filled and reclaimed land 20%
- Cut ground 11%

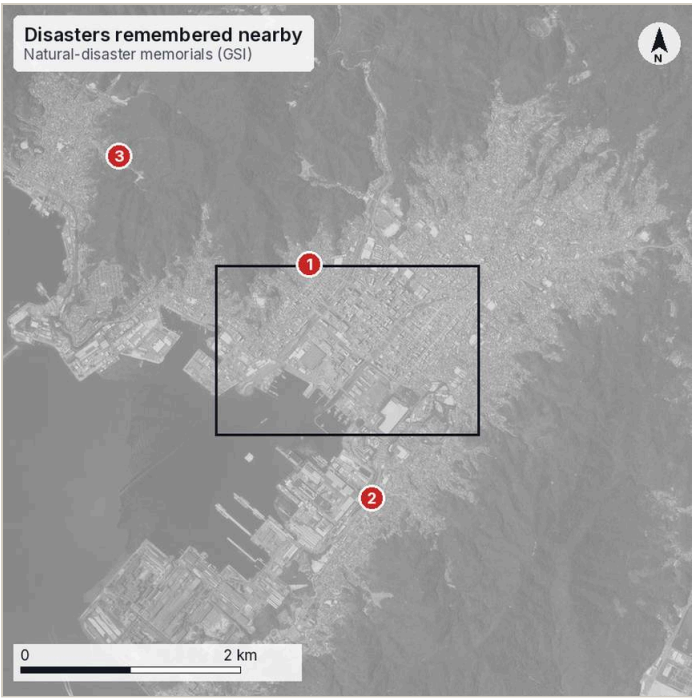
Meiji-era lowland

- River, pond or sea 16%
- Rice paddy 12%
- Tidal flat or beach 6.9%
- Uncultivated land 3.4%
- Marsh 2.7%
- Sand and gravel 1.4%

Filled and reclaimed land (20%). Ground raised with fill, land made by dumping soil into the sea, rivers or ponds, or valleys filled in. May flood if not raised enough; on hills, rain or earthquakes can make the fill fail; in lowlands the liquefaction tendency is very strong, especially on filled sea, lake or river beds.

Cut ground (11%). Ground made by cutting into slopes of mountains or terrace edges. Depending on the cut slope, heavy rain or earthquakes may bring it down; the ground itself is generally firm.

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 6 km across.

Memorial	Disaster	Erected	Distance
1 Monument recording the 2001 Geiyo earthquake 芸予地震伝承碑	Geiyo earthquake (24 Mar 2001) · earthquake	unknown	0.9 km
2 Memorial to the Miyahara victims of the 1945 typhoon and 1967 heavy rain 慰霊碑	Makurazaki Typhoon (17 Sep 1945) and heavy rain of July 1967 (9 Jul 1967) · landslide or debris flow	1995	1.4 km
3 Record of the July 2018 heavy-rain disaster: Yoshiura 1 平成30年7月 豪雨災害の記録 吉浦1	Heavy rain of July 2018 (western Japan floods) · flood / landslide or debris flow	2024	2.7 km

What the memorials record

1. **Monument recording the 2001 Geiyo earthquake.** The Geiyo earthquake at 3:27 pm on 24 March 2001 badly damaged Tainomiya Shrine in Nishimitsuta-machi, Kure: its torii gate collapsed, and roof tiles and the stone guardian dogs (komainu) fell.
2. **Memorial to the Miyahara victims of the 1945 typhoon and 1967 heavy rain.** The Makurazaki Typhoon of 17 September 1945 killed 1,154 people in the then city of Kure and swept away 1,162 houses; 309 of the dead were in Miyahara. On 9 July 1967 torrential rain along the rainy-season front caused landslides and floods across the city, killing 88 people in Kure, 14 of them in Miyahara.
3. **Record of the July 2018 heavy-rain disaster: Yoshiura 1.** In the heavy rain of July 2018, debris flows and slope collapses at many points in the Yoshiura district destroyed buildings and cut roads, and 3 people died in this part of Yoshiura.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo2* (1979–1983), *gazo4* (1987–1990) and *nendophoto2023*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.4 × 1.5 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

Compiled by Japan Time Atlas; not produced, reviewed or endorsed by GSI, MLIT or any Japanese government agency. Not an official hazard map and not for statutory real-estate disclosure. Generated 2026-09-27 by an automated pipeline (research/lab-07/scripts/basic.py).